Capstone Project: Finance Agent - Investment Research Agent

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Document Classification: Confidential
Document Version: 1.0.0
Target Platform: Google Cloud Platform (Vertex AI, Google Agent Development Kit, Cloud Run, BigQuery, Document AI)
Last Major Revision: Aug 14, 2026

Executive Summary & Business Context

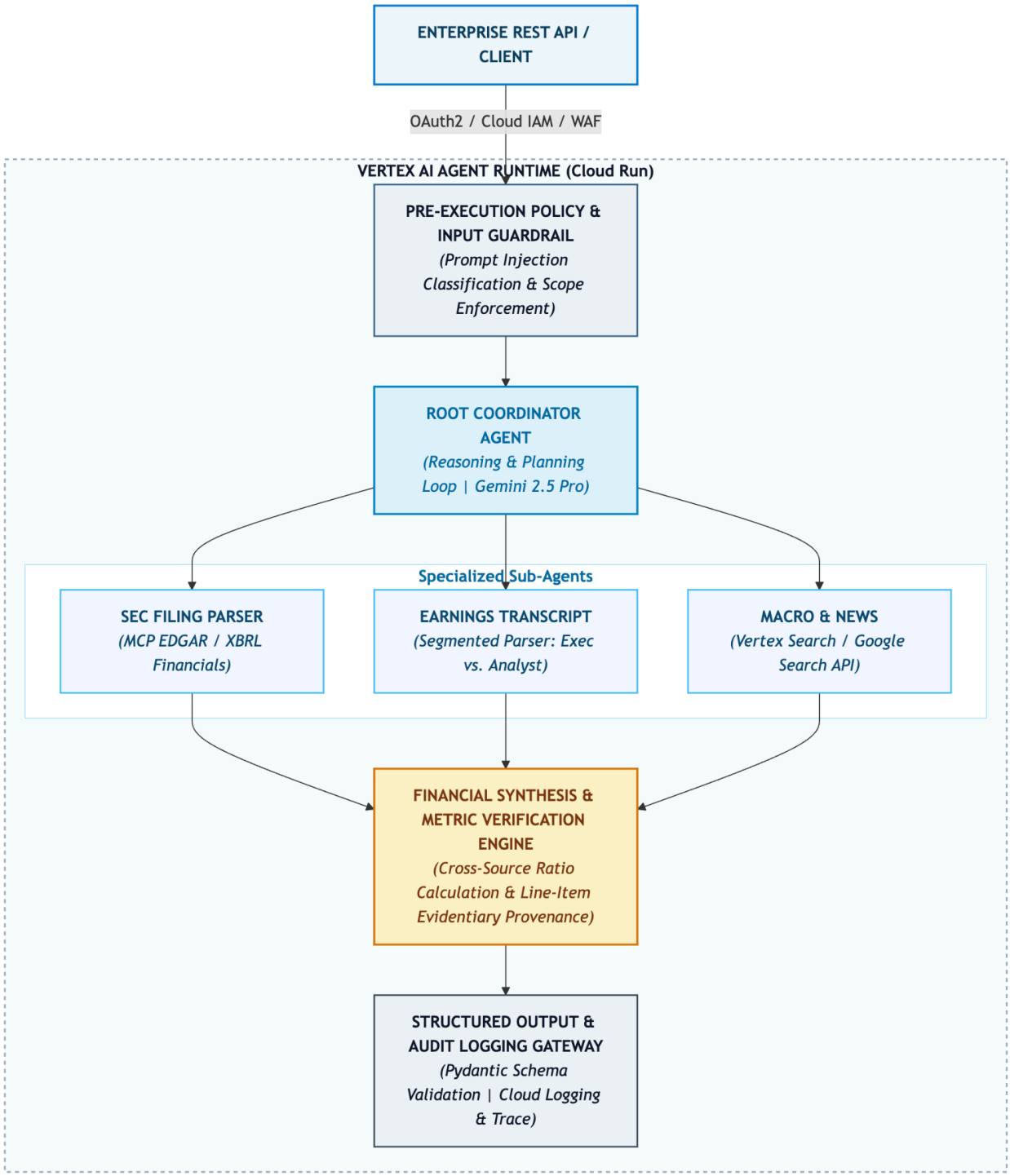
In institutional financial institutions, buy-side asset management firms, and equity research desks, financial analysts spend over **50% of their billable hours** performing manual data aggregation across fragmented corporate repositories. Analysts must search regulatory portals for audited filings (SEC Form 10-K, 10-Q, 8-K), review 20+ page quarterly earnings call transcripts to evaluate executive sentiment, and monitor real-time macroeconomic news feeds.

This manual workflow introduces critical bottlenecks:

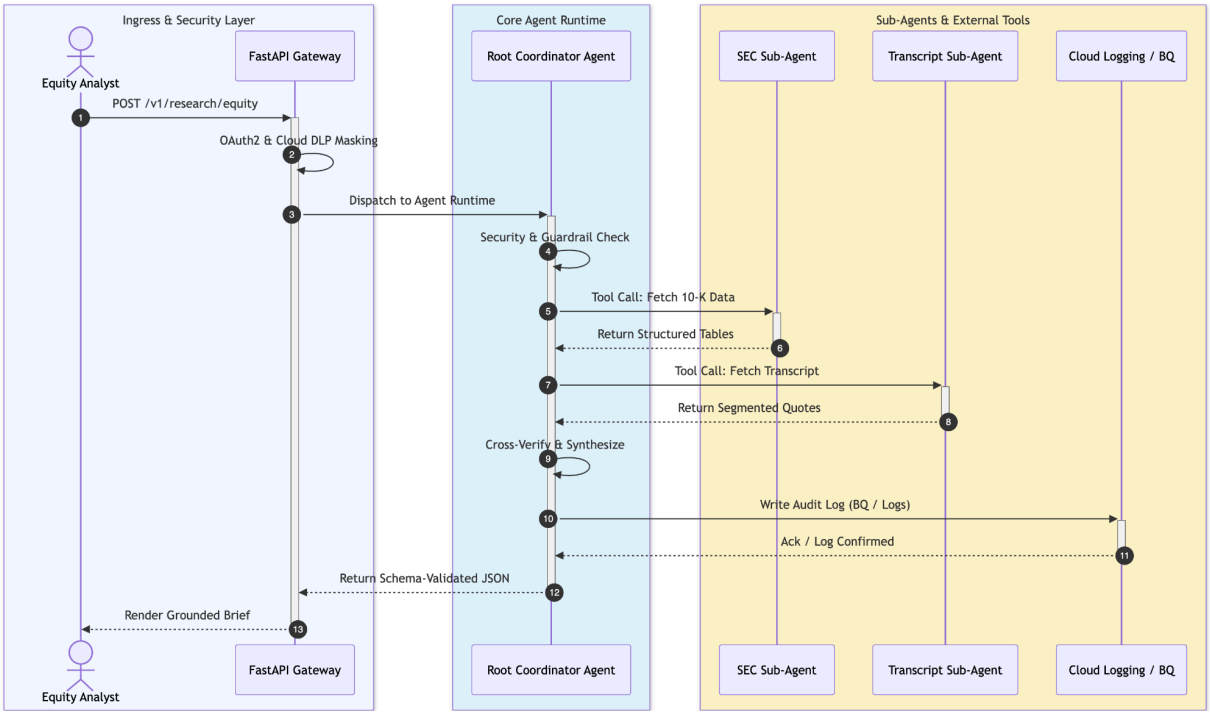
1. **High Synthesis Latency & Operational Drag:** Manually cross-referencing multi-page audited financial statements with qualitative executive commentary takes between 3 to 6 hours per company per quarter.
2. **Hallucination Risk & Missing Provenance in Legacy AI:** General-purpose LLMs frequently hallucinate numerical disclosures or cite stale training data, rendering outputs unviable for institutional compliance and investment committees.
3. **Information Fragmentation:** Analysts must independently navigate SEC EDGAR, third-party transcript aggregators, and live news feeds, leading to missed qualitative risk disclosures in Item 1A.

The **Autonomous Investment Research Intelligence Agent (IRA)** is an enterprise multi-agent system built on Google Cloud. It automates equity due diligence by orchestrating specialized sub-agents across audited regulatory filings, earnings transcripts, and live market intelligence. The system delivers audit-ready financial research briefs with line-item citation provenance in under 60 seconds.

High-Level System Architecture



End-to-End System Interaction Flow



Detailed Sequence Flow Specification

Step	Initiating Component	Target Component	Action & Payload Description
1	Equity Analyst	API Gateway	POST /v1/research/equity with ticker ("AAPL"), fiscal_year (2024), and focus_areas (["Services_Margin", "China_Headwinds"]).
2	API Gateway	API Gateway	Validates OAuth2 token; runs Cloud DLP to redact proprietary portfolio metadata and analyst identifiers.
3	API Gateway	Root Coordinator Agent	Dispatches execution request to the multi-agent runtime with sanitized parameters.

Step	Initiating Component	Target Component	Action & Payload Description
4	Root Coordinator Agent	Input Guardrail	Validates prompt against financial scope policy; blocks prompt injection and out-of-domain jailbreaks.
5	Root Coordinator Agent	SEC Sub-Agent	Invokes tool: fetch_sec_filing_section(ticker="AAPL", form="10-K", section="MD&A").
6	SEC Sub-Agent	Root Coordinator Agent	Returns parsed markdown tables with Balance Sheet, Segment Revenue, and exact XBRL line-item metadata.
7	Root Coordinator Agent	Transcript Sub-Agent	Invokes tool: get_earnings_transcript(ticker="AAPL", quarter="Q4_2024", speaker_role="CFO").
8	Transcript Sub-Agent	Root Coordinator Agent	Returns segmented CFO remarks and analyst Q&A regarding forward gross margin guidance and regional headwinds.
9	Root Coordinator Agent	Synthesis Engine	Cross-verifies quantitative numbers with executive statements; calculates ratio metrics and checks consistency.
10	Root Coordinator Agent	Cloud Logging / BQ	Writes immutable audit log: timestamp, user, ticker, latency_ms, tools_called, token_usage, prompt_hash.
11	Root Coordinator Agent	API Gateway	Returns schema-validated JSON with executive summary, financial tables, and line-item citations.
12	API Gateway	Equity Analyst	Renders comprehensive, audit-ready financial research brief with interactive citation links and risk matrices.

Phase 1: Plan & Discover

Use Case Definition

Objectives

- **Problem Definition:** Articulate the business bottleneck of manual equity research and define clear opportunities for autonomous synthesis.
- **Stakeholder Alignment & Success Criteria:** Establish concrete Definition of Done thresholds (end-to-end latency <= 60s, 100% evidentiary provenance, >= 80% test coverage).
- **Domain-Applied AI/ML Expertise:** Translate financial ratios and qualitative executive guidance into verifiable model evaluation objectives.

Technical Requirements & Scope

- **Problem Formalization:** Document the pain points of fragmented data repositories, synthesis latency, and model hallucination risks.
- **Target Personas & User Journeys:** Detail workflows and acceptance criteria for Institutional Equity Analysts, Portfolio Managers, and Risk/Compliance Leads.
- **Domain KPI Translation:** Define programmatic formulas for financial metrics (YoY Revenue Growth, Operating Margin Compression, Free Cash Flow Conversion, Debt-to-Equity).
- **System Boundaries:** Formulate strict boundary rules distinguishing public regulatory disclosures (in-scope) from trade execution and internal non-public information (out-of-scope).

Deliverable Artifacts

- **Functional Requirements Document (FRD / PRD):** Master specification detailing business context, user personas, baseline vs. target workflow timings, and ROI metrics.
- **Financial Domain KPI Specification:** Document defining mathematical formulas, data sources, and validation thresholds for target financial metrics.

Data & Deploy Discovery

Objectives

- **Technical Scope & Constraints:** Establish ingestion boundaries, data structures, and infrastructure sizing limits.

- **Data Protection & Privacy:** Identify sensitive proprietary data and configure inspection and masking policies.
- **Resource Efficiency:** Size language models and context caches for cost-effective inference (<$0.25 per synthesized brief).

Technical Requirements & Scope

- **Data Ingestion Analysis:** Profile data structures across SEC filings (HTML/XBRL tables), earnings call transcripts, and live market feeds.
- **Data Protection & Privacy Review:** Identify proprietary metadata risks and specify Cloud DLP inspection rules to mask client identifiers.
- **Deployment Target Discovery:** Determine runtime constraints for deployment on Google Cloud Run (2 vCPU, 4GB RAM, max 50 instances, concurrency 80) within a private VPC.

Deliverable Artifacts

- **Data & Deployment Discovery Document:** Technical specification detailing data schemas, ingestion volumes, rate-limit constraints, and Cloud Run runtime parameters.
- **Data Dictionary & Schema Specification:** Catalog of SEC table structures, transcript speaker labels, and metadata tags.

Grounding & API Discovery

Objectives

- **Retrieval & Data Engineering for AI:** Assess external data interfaces, embedding patterns, and tool connectivity protocols.
- **Modularity & Abstraction:** Decouple tool interfaces using standardized communication protocols (MCP / OpenAPI).
- **API Design & Versioning:** Enforce contract-first design with backward-compatible schemas.

Technical Requirements & Scope

- **Tool & API Protocol Evaluation:** Evaluate integration options for upstream data sources:
    - Official SEC EDGAR Submissions API and Model Context Protocol (MCP) tool standard.
    - Transcript retrieval endpoints with speaker role parsing.
    - Real-time search grounding via Vertex AI Search and Google Search Grounding API.

- **Contract-First API Design:** Draft the external service interface conforming to the OpenAPI 3.1 specification, establishing versioned endpoints (/v1/research/equity, /v1/research/filings).
- **Rate-Limit & Backoff Discovery:** Document upstream API rate limits (e.g., SEC EDGAR limit of 10 requests/sec with mandatory User-Agent headers) and formulate retry strategies.

Deliverable Artifacts

- **Grounding & API Discovery Document:** Architecture analysis of external data sources, grounding mechanisms, and MCP protocol integration contracts.
- **OpenAPI REST Specification:** Standardized API contract specifying request/response payloads, error schemas, and security schemes.

Golden Dataset

Objectives

- **LLM Ops and Evaluation:** Create a comprehensive ground-truth test suite for regression testing and accuracy benchmarking.
- **Testing & Quality Engineering:** Establish quantitative acceptance criteria across diverse financial edge cases.

Technical Requirements & Scope

- **Curated Evaluation Benchmark:** Construct a golden benchmark dataset containing **50+ diverse financial evaluation scenarios** spanning multiple market sectors (Technology, Healthcare, Energy, Financials).
- **Multi-Turn & Edge-Case Coverage:** Ensure dataset includes:
    - Standard extraction queries (e.g., segmented revenue breakdown).
    - Complex comparative queries across fiscal years (e.g., margin compression vs. CFO forward commentary).
    - Ambiguous queries requiring multi-agent decomposition.
    - Adversarial / Out-of-Scope queries (e.g., prompt injection, code generation requests, stock price predictions).
- **Ground-Truth Labeling:** Provide verified, human-annotated ground truth for every scenario, including exact primary-source filing citations and expected numerical figures.

Deliverable Artifacts

- **Golden Evaluation Dataset:** Structured benchmark dataset containing input prompts, reference documents, ground-truth answers, and test tags.

- **Dataset Curation & Annotation Guidelines:** Documentation detailing the methodology for dataset creation, source verification, and ongoing benchmark expansion.

Phase 2: Architecture

Solution Architecture

Objectives

- **System Design Artifacts:** Create formal block diagrams, data flow models, and interaction sequence diagrams.
- **Decision Records (ADRs):** Document technical trade-offs, architecture alternatives, and rationale.
- **Agentic & Multi-Agent Systems:** Define coordination topologies, planner reasoning loops, and state management.
- **Availability Design & Graceful Degradation:** Architect high-availability services with circuit breakers and structured fallbacks.
- **AI-Specific Security:** Model threat vectors including prompt injection and tool execution boundaries.

Technical Requirements & Scope

- **Multi-Agent Topology Design:** Design a hierarchical multi-agent architecture using the Google Agent Development Kit (ADK), establishing clean separation between the Root Coordinator Agent and domain-specific sub-agents.
- **Reasoning Loop & Planner Design:** Specify the planner execution loop (Plan-and-Execute / ReAct) with explicit thought budgeting in Gemini 2.5 Pro.
- **Architecture Decision Records (ADRs):** Formulate 3 comprehensive ADRs:
    1. ADR-001: Multi-Agent Hierarchical Orchestration (ADK) vs. Monolithic ReAct Loop.
    2. ADR-002: Model Context Protocol (MCP) Tool Integration vs. Proprietary REST Wrappers.
    3. ADR-003: Gemini Context Caching vs. Chunk-based Vector Search for 150+ page annual 10-K Filings.
- **Sequence & Threat Modeling Artifacts:** Produce architectural block diagrams, end-to-end sequence diagrams, and threat models covering prompt injection, tool tampering, and data exfiltration.

Deliverable Artifacts

- **Technical Design Document (TDD) — Part 1 (System Architecture):** Master technical design covering system topology, component interactions, sequence flows, data contracts, and fault tolerance.

- **Architecture & Data Flow Diagrams:** Visual system block diagrams, component topology, and UML sequence diagrams.
- **Architecture Decision Records (ADRs):** Formal ADR documents detailing problem context, options considered, trade-offs, and final decisions.

App Architecture & Scaffold

Objectives

- **Modularity & Abstraction:** Implement clean interfaces and dependency injection patterns for swappable components.
- **Configuration Management:** Decouple application code from runtime environments, credentials, and model prompts.
- **Extensibility:** Establish plugin and event-driven patterns for future data source onboarding.
- **Code Quality & AI Lifecycle Management:** Implement strict linting, type safety, and versioned prompt assets.

Technical Requirements & Scope

- **Application Scaffolding:** Create a modular Python repository structure using FastAPI and Pydantic v2.
- **Interface Abstractions:** Define abstract base classes and interfaces (BaseAgent, BaseTool, FilingRetrievalInterface, MarketDataInterface) using dependency injection to allow swappable backend providers.
- **Configuration Management:** Externalize runtime settings (model identifiers, temperature, token limits, safety thresholds) into environment variables and versioned configuration files.
- **Developer Tooling & Linting:** Establish code quality tooling with automated linting, formatting, type checking, and pre-commit hooks.

Deliverable Artifacts

- **Technical Design Document (TDD) — Part 2 (Application Architecture):** Application design specification detailing module structures, interface contracts, and dependency injection patterns.
- **Application Framework & Scaffolding Code:** Modular application skeleton, base interface classes, and middleware setup.
- **Agent & Prompt Configuration Files:** Externalized configuration files managing system instructions, few-shot examples, and model parameters.
- **Data Models & Schema Code:** Pydantic data models defining domain entities, tool parameters, and response schemas.

Infra & Datastore Provision

Objectives

- **Infrastructure as Code (IaC):** Codify 100% of cloud resources in declarative, reproducible configuration scripts.
- **Infrastructure & Network Security:** Enforce private network perimeters, VPC peering, and secure ingress/egress.
- **Authentication & Authorization:** Configure fine-grained IAM service accounts following the principle of least privilege.
- **Retrieval & Data Engineering for AI:** Provision scalable vector indexing endpoints and analytical datastores.

Technical Requirements & Scope

- **100% Infrastructure as Code (Terraform):** Author modular Terraform scripts provisioning all required Google Cloud infrastructure.
- **Network & Compute Isolation:** Provision a dedicated Virtual Private Cloud (VPC), Private Google Access, Serverless VPC Access Connectors, and Cloud Run service configs.
- **Datastore & Vector Search Provisioning:** Provision **Vertex AI Vector Search** Index and Endpoint, **BigQuery** analytics datasets, and secure **Google Cloud Storage (GCS)** buckets.
- **Security & KMS Infrastructure:** Configure Google Secret Manager, Cloud KMS customer-managed encryption keys, and fine-grained IAM service accounts following the principle of least privilege.

Deliverable Artifacts

- **Infrastructure as Code (Terraform) Modules:** Complete Terraform configurations provisioning VPC, Cloud Run, Vertex AI Vector Search, BigQuery, and Secret Manager.
- **IAM & Security Policy Code:** Role definitions, service account assignments, and least-privilege access policies.
- **Database DDL & Schema Scripts:** SQL table definitions for financial time-series metrics, ratio analytics, and compliance audit storage.

Phase 3: Build & Evaluate

ADK Code Development

Objectives

- **Agentic & Multi-Agent Systems:** Implement autonomous planning, multi-agent dispatch, and stateful synthesis.
- **Retrieval & Data Engineering for AI:** Build table-aware parsing, hybrid search, and context caching pipelines.
- **Model Selection, Tuning & Optimization:** Enforce structured outputs and optimize model tiers across agents.
- **AI-Specific Security:** Build pre-execution input guardrails and policy classification filters.
- **Graceful Degradation:** Implement resilient fallbacks and timeout handling during tool execution.

Technical Requirements & Scope

- **Root Coordinator Agent:** Build the central orchestrator using Google ADK and Gemini 2.5 Pro (with Thinking Budget), implementing reasoning loops to decompose complex queries and synthesize multi-agent outputs.
- **Specialized Sub-Agents:**
    - **SEC Filing Specialist:** Ingests and parses Form 10-K/10-Q/8-K documents, extracting structured balance sheets, income statements, and MD&A sections via the SEC EDGAR MCP Server.
    - **Earnings Transcript Specialist:** Parses quarterly call transcripts, separating formal executive prepared remarks from analyst Q&A.
    - **Macro News Specialist:** Executes grounded web queries using Vertex AI Search to capture real-time interest rate decisions and industry sentiment.
- **Context Caching Layer:** Implement Gemini Context Caching on static annual 10-K filings to minimize repetitive prompt processing costs.
- **Deterministic Financial Verification Tools:** Author Python execution tools computing core financial ratios (growth, margins, leverage) directly from primary figures.
- **Pre-Execution Guardrails:** Build pre-model input guardrails running classifier checks to intercept prompt injection attempts and reject out-of-domain queries.

Deliverable Artifacts

- **Multi-Agent Core Application Code:** Python codebase implementing Root Coordinator, specialized sub-agents, reasoning planner, and report synthesis engine.

- **MCP Server & Tool Integration Code:** Model Context Protocol (MCP) client integrations and custom financial calculator tools.
- **Security Guardrails Code:** Pre-execution classifier callbacks enforcing financial scope and blocking prompt injections.
- **Context Caching & Ingestion Code:** Data processing pipelines managing Gemini context caches and vector store embeddings.

ADK Evaluation (Eval Run)

Objectives

- **LLM Ops and Evaluation:** Quantify agent performance using automated evaluation metrics (Faithfulness, Relevance, Citation Precision).
- **AI Lifecycle Management:** Track experiment runs, prompt variations, and model parameter changes.
- **Testing & Quality Engineering:** Establish automated regression gates asserting quality thresholds.

Technical Requirements & Scope

- **Automated Evaluation Pipeline:** Build an automated evaluation suite using **Vertex AI Gen AI Evaluation** assessing agent outputs against the Golden Dataset.
- **Quantitative Quality Metrics:**
    - **Faithfulness / Groundedness:** Target score of **1.0** (zero hallucination permitted on financial numbers).
    - **Answer Relevance & Completeness:** Target score of **>= 0.90** against reference benchmarks.
    - **Citation Precision:** Target **100%** verifiable source alignment (filing type, fiscal period, exact row/column or quote).
- **Automated Regression Test Suite:** Implement evaluation regression testing to automatically detect quality regressions upon prompt or model parameter updates.

Deliverable Artifacts

- **Evaluation Pipeline Code:** Automated benchmark execution scripts calculating Faithfulness, Relevance, and Citation metrics.
- **Evaluation Baseline Report:** Documented benchmark analysis detailing metric score distributions, latency statistics, and error categorization.
- **Regression Test Automation Scripts:** Quality gate scripts integrated into testing workflows to assert threshold compliance.

Phase 4: Deploy & Verify

Deploy & Publish

Objectives

- **CI/CD & Deployment:** Automate build verification, container packaging, evaluation gates, and zero-downtime rollouts.
- **Scalability & Elasticity:** Configure horizontal autoscaling policies to accommodate query spikes.
- **AI Cost Management:** Optimize token usage and model routing to control operational expenses.

Technical Requirements & Scope

- **CI/CD Pipeline (Cloud Build):** Author a declarative pipeline automating:
    1. Code linting and static type checks.
    2. Unit and integration test execution.
    3. Evaluation benchmark gating (blocking builds on metric regression).
    4. Container packaging and publishing to Google Artifact Registry.
    5. Zero-downtime blue/green deployment to Google Cloud Run.
- **Cloud Run Production Deployment:** Deploy the containerized agent gateway with horizontal autoscaling (0 to 50 instances) and concurrency set to 80 requests per container instance.
- **Cost & Token Optimization:** Implement model routing (delegating parsing to Gemini 2.5 Flash, reserving Gemini 2.5 Pro for synthesis) to ensure total per-query cost stays within target budget caps.

Deliverable Artifacts

- **CI/CD Build Pipeline Configurations:** Declarative build scripts defining testing, evaluation gating, and container deployment stages.
- **Container Build Specifications:** Production-ready, security-hardened container build definitions.
- **Cloud Deployment Configuration:** Infrastructure configurations managing Cloud Run traffic splits, CPU/memory allocations, and autoscaling thresholds.

Integration Testing

Objectives

- **Testing & Quality Engineering:** Validate end-to-end user workflows and maintain high test coverage (>= 80%).
- **Failure & Recovery Testing:** Conduct chaos tests simulating upstream outages, rate limits, and latency spikes.
- **Graceful Degradation:** Verify that system fallbacks maintain partial service when non-critical components fail.

Technical Requirements & Scope

- **End-to-End Integration Tests:** Execute comprehensive integration test suites verifying API gateway authentication, multi-agent tool execution, and database logging.
- **Upstream Failure & Chaos Testing:** Perform automated failure injection testing simulating:
    - Upstream SEC EDGAR 429 Rate Limit responses.
    - Vertex AI model endpoint 503 transient errors.
    - Real-time news search network timeouts.
- **Resilience & Fallback Validation:** Verify that exponential backoff with jitter operates correctly and that circuit breakers gracefully fall back to cached historical filings without failing the user request.
- **Code Coverage Requirement:** Enforce a minimum of **>= 80% code coverage** across all application modules.

Deliverable Artifacts

- **Integration & E2E Test Suite Code:** Automated test scripts verifying end-to-end multi-agent execution and external tool contracts.
- **Chaos & Failure Injection Test Scripts:** Automated resilience test scripts simulating upstream API timeouts and network partitions.
- **Test Execution & Code Coverage Reports:** Formatted test results and coverage reports documenting compliance with quality gates.

Ecosystem Verification

Objectives

- **Observability:** Instrument distributed tracing, latency metrics, and operational dashboards.

- **Compliance & Governance:** Capture tamper-evident audit logs for regulatory oversight and tracing.
- **Extensibility:** Integrate event-driven alerting channels for downstream consumers.

Technical Requirements & Scope

- **Distributed Tracing (OpenTelemetry):** Instrument full-stack OpenTelemetry tracing across the FastAPI gateway, agent planner, tool invocations, and Vertex AI API calls, exporting spans to Google Cloud Trace.
- **Structured Audit Logging:** Implement structured compliance logging in Google Cloud Logging, recording every query with: `timestamp`, `user_id`, `input_prompt_fingerprint`, `tools_called`, `model_id`, `latency_ms`, and `safety_verdict`.
- **SLO Telemetry & Dashboards:** Construct a Cloud Monitoring dashboard tracking Service Level Objectives (99.9% availability, P95 latency <= 60s, error rate < 2%).
- **Event-Driven Extensibility:** Configure Google Cloud Pub/Sub integration to broadcast completed research brief events and critical risk signals to downstream messaging webhooks and dashboards.

Deliverable Artifacts

- **Observability & Tracing Instrumentation Code:** Distributed tracing setup exporting latency spans to Cloud Trace.
- **Structured Audit Logging Middleware:** Compliance audit logger streaming tamper-evident execution records to Cloud Logging and BigQuery.
- **Cloud Monitoring Dashboard Definitions:** Dashboard specifications visualizing system health, token usage, latency percentiles, and error rates.
- **Event-Driven Notification Code:** Asynchronous message publisher for Cloud Pub/Sub event broadcasting.

Manual Guide

Objectives

- **Operational Documentation:** Author comprehensive runbooks for maintenance, secret rotation, and incident triage.
- **API Documentation:** Provide clear, interactive developer guides and query walkthroughs.

Technical Requirements & Scope

- **Operational Runbook:** Author an operational Runbook detailing daily maintenance procedures, Secret Manager key rotation, upstream API credential updates, and incident triage workflows.
- **Developer Setup Guide:** Document local sandbox setup, environment variable requirements, pre-commit configuration, and testing commands.
- **User & API Demonstration Guide:** Provide interactive user documentation with sample financial research queries, expected JSON outputs, citation inspection steps, and error-handling explanations.

Deliverable Artifacts

- **Operational Runbook:** Comprehensive operational guide for system maintenance, failure recovery, key rotation, and on-call troubleshooting.
- **Developer Onboarding & Setup Guide:** Instructions for local development environment configuration, testing, and toolchains.
- **User & API Demonstration Walkthrough:** Step-by-step demonstration guide detailing query workflows, expected outputs, and citation verification.

Final Report

Objectives

- **Stakeholder Alignment & Success Criteria:** Validate and report on final business and technical outcomes.
- **AI Cost Management & Forecasting:** Present telemetry on token costs, caching efficiency, and cloud spend.
- **Architecture Documentation & Communication:** Document architectural maturity, trade-offs, and future roadmap.

Technical Requirements & Scope

- **Product Readiness Review (PRR) Readout:** Synthesize overall project outcomes into an executive-level PRR presentation deck validating that all success criteria have been satisfied.
- **Financial ROI & Cost Analysis:** Document comprehensive cost telemetry, breaking down average token consumption per query, Gemini Context Caching savings, and estimated cloud hosting costs per 1,000 queries.
- **Architectural Roadmap & Next Steps:** Outline future enhancements (e.g., expanding coverage to international filings, automated audio earnings call ingestion, proactive morning alerts).

Deliverable Artifacts

- **Final Technical & Business Summary Report:** Comprehensive project closure report documenting architecture, evaluation benchmarks, and business outcomes.
- **Product Readiness Review (PRR) Presentation Deck:** Executive slide deck summarizing system capabilities, compliance reviews, and operational readiness.
- **Cost & Token Consumption Analysis Report:** Financial cost model detailing token usage, context caching savings, and infrastructure expenditure forecasts.

Objective Mapping Matrix

ADLC Phase	ADLC Stage	Primary Objectives	Key Deliverables
Phase 1: Plan & Discover	**Use Case Definition**	Scoping & Documentation (Problem Definition), AI/ML (Domain Expertise)	Functional Requirements Document (FRD), Financial KPI Specification
	Data & Deploy Discovery	Scoping (Constraints), Security (Data Protection), Performance	Data & Deploy Discovery Document, Data Dictionary
	Grounding & API Discovery	AI/ML (Retrieval & Data Engineering), Designing for Change (API Design, Modularity)	Grounding & API Discovery Document, OpenAPI REST Specification
	Golden Dataset	AI/ML (LLM Ops & Evaluation), Operational Excellence (Testing & Quality)	Golden Evaluation Dataset, Dataset Annotation Guidelines
Phase 2: Architecture	**Solution Architecture**	Scoping (Design Artifacts, ADRs), AI/ML (Agentic Systems), Reliability (Availability)	Technical Design Document (Part 1), Architecture Diagrams, Architecture Decision Records (ADRs)
	App Architecture & Scaffold	Designing for Change (Modularity, Config), Ops (Code Quality, AI Lifecycle)	Technical Design Document (Part 2), Application Scaffold Code, Configuration Files, Data Model Code
	Infra & Datastore Provision	Ops (IaC / Terraform), Security (Network Security, IAM), AI/ML (Data Engineering)	Infrastructure as Code Modules, IAM Policies, Database DDL Scripts
Phase 3: Build & Evaluate	**ADK Code Development**	AI/ML (Agentic Systems, RAG, Optimization), Security (Guardrails), Reliability	Multi-Agent Application Code, MCP Integration Code, Security Guardrails Code, Caching Code

ADLC Phase	ADLC Stage	Primary Objectives	Key Deliverables
	ADK Evaluation (eval run)	AI/ML (LLM Ops & Evaluation), Ops (AI Lifecycle, Quality Engineering)	Evaluation Pipeline Code, Evaluation Baseline Report, Regression Test Scripts
Phase 4: Deploy & Verify	**Deploy & Publish**	Ops (CI/CD & Deployment), Performance (Scalability, Cost Management)	CI/CD Pipeline Configurations, Container Build Specs, Deployment Configurations
	Integration Testing	Ops (Testing & Quality), Reliability (Failure & Recovery Testing, Degradation)	Integration Test Suite Code, Chaos Test Scripts, Code Coverage Reports
	Ecosystem Verification	Reliability (Observability), Security (Audit Logging), Designing for Change (Extensibility)	Tracing Instrumentation Code, Audit Logging Middleware, Dashboard Definitions, Notification Code
	Manual Guide	Scoping & Documentation (Operational Documentation, API Documentation)	Operational Runbook, Developer Setup Guide, User & Demo Walkthrough Guide
	Final Report	Scoping (Success Criteria), Performance (AI Cost Management), Architecture Communication	Final Project Report, Product Readiness Review Presentation Deck, Cost Analysis Report

Consolidated Deliverable Artifacts List

1. Requirements & Discovery Documents

- **Functional Requirements Document (FRD / PRD):** Problem definition, user personas, baseline vs. target workflow timings, and ROI metrics.
- **Financial Domain KPI Specification:** Formulas, data sources, and validation rules for calculated ratios.
- **Data & Deployment Discovery Document:** Ingestion data schemas, volume constraints, and Cloud Run runtime parameters.
- **Grounding & API Discovery Document:** Evaluation of upstream SEC EDGAR API, MCP protocol, and search grounding.
- **Data Dictionary & Schema Specification:** Catalog of SEC tables, transcript speaker labels, and metadata tags.

2. Architecture & Technical Design Documents (TDD)

- **Technical Design Document (TDD):** Comprehensive system architecture covering component topology, sequence flows, data contracts, and fault tolerance.
- **Architecture Decision Records (ADRs):** Formal records for Multi-Agent Orchestration (ADR-001), MCP Protocol Integration (ADR-002), and Context Caching vs. Vector RAG (ADR-003).
- **Architecture & Data Flow Diagrams:** Visual system block diagrams, component topology, and UML sequence diagrams.
- **OpenAPI REST Specification:** Standardized OpenAPI 3.1 contract for /v1/research/equity and /v1/research/filings.

3. Application Source Code & Configuration

- **Multi-Agent Core Application Code:** Python codebase containing the Root Coordinator Agent, specialized sub-agents (SEC, Transcripts, Macro News), and synthesis engine.
- **MCP Server & Tool Integration Code:** Model Context Protocol (MCP) clients and financial verification tools.
- **Security Guardrails Code:** Pre-execution classifier callbacks for prompt injection mitigation and scope enforcement.
- **Context Caching & Ingestion Code:** Data processing pipelines for Gemini context caching and vector indexing.
- **Agent & Prompt Configuration Files:** Version-controlled YAML files managing system instructions, few-shot prompts, and model parameters.

- **Data Models & Schema Code:** Pydantic data models for structured request/response validation.

4. Infrastructure as Code (IaC) & CI/CD

- **Infrastructure as Code (Terraform) Modules:** Declarative Terraform scripts provisioning VPC, Cloud Run, Vertex AI Vector Search, BigQuery, GCS, and Secret Manager.
- **IAM & Security Policy Code:** Service account configurations and least-privilege role bindings.
- **CI/CD Build Pipeline Configurations:** Declarative build files automating linting, testing, eval gates, and zero-downtime deployment.
- **Container Build Specifications:** Production-ready, multi-stage Docker container specifications.
- **Database DDL & Schema Scripts:** SQL table definitions for financial time-series metrics and audit logging.

5. Datasets, Testing & Evaluation Artifacts

- **Golden Evaluation Dataset:** Curated benchmark containing 50+ diverse financial evaluation scenarios with ground-truth labels.
- **Dataset Curation & Annotation Guidelines:** Documentation of dataset curation methodology and source verification rules.
- **Evaluation Pipeline Code:** Automated benchmark execution scripts computing Faithfulness, Relevance, and Citation Precision.
- **Evaluation Baseline Report:** Detailed analysis of baseline evaluation scores, latency distributions, and failure modes.
- **Integration & E2E Test Suite Code:** Automated unit, integration, and regression test suites maintaining >= 80% coverage.
- **Chaos & Failure Injection Test Scripts:** Automated resilience test scripts simulating upstream API outages and rate limits.
- **Test Execution & Code Coverage Reports:** Formatted test results and coverage reports.

6. Observability, Security & Operations

- **Observability & Tracing Instrumentation Code:** OpenTelemetry setup exporting distributed traces to Cloud Trace.
- **Structured Audit Logging Middleware:** Compliance audit logger streaming tamper-evident execution records to Cloud Logging and BigQuery.
- **Cloud Monitoring Dashboard Definitions:** Dashboard specifications visualizing system health, token usage, latency percentiles, and error rates.

- **Event-Driven Notification Code:** Asynchronous message publisher for Cloud Pub/Sub event broadcasting.
- **Operational Runbook:** Comprehensive operational guide for system maintenance, failure recovery, key rotation, and on-call troubleshooting.
- **Developer Onboarding & Setup Guide:** Instructions for local development environment configuration, testing, and toolchains.
- **User & API Demonstration Walkthrough:** Step-by-step demonstration guide detailing query workflows, expected outputs, and citation verification.

7. Final Project Reports & Executive Readouts

- **Final Technical & Business Summary Report:** Comprehensive project closure report documenting architecture, evaluation benchmarks, and business outcomes.
- **Product Readiness Review (PRR) Presentation Deck:** Executive slide deck summarizing system capabilities, compliance reviews, and operational readiness.
- **Cost & Token Consumption Analysis Report:** Financial cost model detailing token usage, context caching savings, and infrastructure expenditure forecasts.

Verifiable Outcomes & Definition of Done

1. Functional Verification

- **Groundedness & Provenance:** 100% of numerical figures, segment revenues, and margin percentages cited in synthesized briefs must match primary SEC Form 10-K/10-Q disclosures or transcript text with explicit line-item citations.
- **Schema Validation:** 100% of API responses must conform strictly to Pydantic schemas without unparsed markdown blocks or invalid JSON structures.
- **Test Coverage:** Application test suites must achieve **>= 80% code coverage** across unit, integration, and guardrail test modules.

2. Operational & Infrastructure Verification

- **Zero-Touch IaC Deployment:** 100% of Google Cloud infrastructure must be provisioned reproducibly via automated Infrastructure as Code scripts with zero manual Cloud Console configuration.
- **Continuous Delivery Pipeline:** Automated CI/CD pipelines must execute linting, unit testing, evaluation benchmarks, and zero-downtime rolling deployments to Cloud Run upon pull request merges.

3. Observability, Security & Latency Verification

- **Latency SLO:** End-to-end multi-source research brief synthesis must complete in **<= 60 seconds** at the 95th percentile (P95).
- **Audit & Security Logging:** 100% of interactive requests, tool executions, token metrics, and guardrail verdicts must be recorded as structured logs in Cloud Logging.
- **Guardrail Interception:** 100% of adversarial jailbreak and prompt injection attempts must be intercepted at the pre-execution boundary before tool dispatch.